Synergistic maturity group and planting date adaptation sustains irrigated soybean yields under warming

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Plain Language Summary

As part of the submission process, we ask authors to prepare a plain language summary. A ninth-grade reading comprehension level is suggested for ease of reading. We suggest you structure your plain language summary with four key elements: subject overview, research purpose, key results, and key takeaways. See detailed instructions here.

The purpose of the plain language summary is to explain the article's research and relevance in clear language free from jargon. If the article is accepted, the summary will appear with the abstract and may be used to further promote your article.

Warmer summers can decrease U.S. soybean seed yields if agronomic management is not adapted accordingly. Our research question is: Can farmers sustain soybean yields under high temperatures by changing cultivar maturity group and sowing dates? We quantified the impacts of high temperatures on soybean yields across thousands of fields in the US Mid-South and investigated whether adapting sowing dates and maturity groups could offset these negative impacts. Soybean yields dropped by nearly 6% on average with a +2Â°C temperature increase. However, planting earlier and choosing a later-maturing soybean variety can increase seed yields by 12% on average under high temperatures. Maturity group selection and planting date are among the most actionable management decisions for farmers, as they can rapidly adopt these changes without incurring significant production costs. These choices can be further optimized under warming to offset yield penalties.

Response to the reviewer and editor comments

We want to thank the reviewers and editor for all the suggestions. All the comments have been carefully analyzed and answered one by one below (our responses are in blue).

Associate Editor's Comments to Author:

One of the reviewers noted that "This revised version is considerably better than the previous one" while another reviewer noted that "the paper is short with minimal content... perhaps work to expand the message to increase the novelty of the paper". It seems the authors have considerably improved the manuscript but the short word limit of this journal may prevent expansion. The reviewers did provide a few comments that could further improve the manuscript. As a last note, both reviewers provided a comment such as "the authors did not provide a point-by-point response to the reviewers' comments, which should have been included" (either in comments to author or comments to editor). While a response to reviewers was provided, it may have been difficult to find given both reviewers mentioned this. Overall, I recommend revisions considering the reviewer comments.
Response: We appreciate the Editor's feedback. We have addressed all the reviewer comments one by one (please see below).

Reviewer 1:

Comment #1
This appears to be a resubmission after the manuscript was originally rejected earlier this year. Overall, my opinion has not changed substantially from my previous reading. The authors have conducted a simulation study to evaluate impacts of earlier planting and variety selection for soybean in the Mid-South to offset heat and drought stress. The paper is short with minimal content. Similar results and conclusions have been found in other literature, so the novelty and impact of the paper is minimal. Authors could perhaps work to expand the message to increase the novelty of the paper. There is plenty of room for expansion to a fuller, more comprehensive research paper.
Response: Thank you for the feedback. While a response to the reviewers was provided addressing this comment in the previous review, it seems the reviewers were unable to find the review letter. Apologies for the inconvenience. As stated in the previous review letter, we believe our contribution is original for the following reasons:

- To our knowledge, this is the first study to quantify the synergistic impact of maturity group and sowing date under possible warming scenarios.
- Our study is unique in terms of resolution of simulations, quantifying impacts across over 4,000 fields and 40 weather years per field.
- We advanced previous simulation studies by exploring stacked management options under warming.

- We have tested and applied the newest APSIM framework (APSIM Next Generation), which advances previous APSIM classic work.

All these points have been clarified and incorporated into the previous review. Lastly, while the reviewer states that there is room for expansion, we have focused on the most important study messages within the recommended word limit (2,500 words) as per the journal guidelines.

Comment #2
L34 – limit
Response: Addressed.

Comment #3
L37 – Subscript the '2'
Response: Addressed.

Comment #4
L47 - Minimize use of first person pronouns here and throughout.
Response: Addressed. The use of first-person pronouns was considerably reduced throughout the whole manuscript.

Comment #5
L62 – model was previously successful
Response: Addressed.

Comment #6
L81 – Define [CO2] on first use.
Response: Addressed.

Comment #7
L95 – Does this mean 2°C was added to max and min T weather station data on a daily basis?
Response: Addressed. The new sentence reads:

"For the high-temperature scenario, 2°C was added to the daily maximum and minimum temperatures from the historical weather station data."

Comment #8
L98 – What is the meaning of "simple"? Simple as compared to what?
Response: Simpler compared to using global circulation climate models for the temperature increase projections, which is also a common approach found in the literature. We have clarified the sentence. It reads:

"The +2°C scenario is a simpler approach compared to using direct outputs of global circulation climate models, but it has produced results with similar direction and magnitude while substantially reducing computational requirements (Elli et al., 2022)."

Comment #9
L100 – How does this reduce computational requirements? The model still needs to run for either case, which likely consumes most of the computational requirement.
Response: Thank you for the question. When global circulation models (GCMs) are used, multiple GCMs and representative concentration pathway scenarios are run, which substantially increases computational requirements compared to the approach adopted in the present study. For example, if at least 15 climate models are run, we would need to run 15 times the 4,651 fields across the 40 weather years considered in the present study, while still providing similar results in direction and magnitude. This was the rationale for using the simpler +2°C scenario.

Comment #10
L109 – What is a site? Is it a pixel that intersects soybean fields in CDL?
Response: Thank you for the suggestion, and we have clarified it. The "site" is interchangeable with "field". In the Materials and Methods section, the following sentence clarifies the definition of a field. It reads:

"The soybean cropland data layer (NASS, 2025) was used to identify soybean-cropped pixels and define the spatial extent of the simulation fields, which were resampled to a 2.5 km grid (~0.025°). Each grid cell classified as soybean cropland was treated as one simulation field."

To avoid confusion, we have replaced "site" with field here and elsewhere.

Comment #11
L115 – were – Use past tense throughout.
Response: Addressed and thank you.

Comment #12
L174-175 – Use past tense.
Response: Addressed and thank you.

Comment #13
L199 – evidence from irrigated soybean in the US Mid-South…
Response: Addressed and thank you.

Reviewer 2:

Comment #14
This revised version is considerably better than the previous one. However, the authors did not provide a point-by-point response to the reviewers' comments, which should have been included. This would have facilitated the review process, particularly since the manuscript was previously submitted to and rejected by this journal.
Response: Thank you for the feedback. We apologize for the inconvenience of not finding a point-by-point response to the previous reviewer's comments. We did submit a review letter addressing all the previous comments, one by one. Apparently, this may have been difficult for the reviewers to find in the journal system.

Comment #15

The main comments raised by this reviewer have been addressed. Although the study has limited novelty and does not provide a mechanistic analysis of the physiological processes underlying the simulated yield responses, it highlights the importance of the interaction between planting date and maturity group and reinforces the need to improve the prediction of these responses under future climate conditions.

Response: Thank you for the positive feedback.

Comment #16

Because the study is limited to an irrigated production system in the Arkansas Delta region, this should be stated more clearly in the title, Introduction, and Discussion. The conclusions may not be directly applicable to rainfed soybean production, which is also common in other parts of the region.

Response: Thank you for the thoughtful suggestion. We have clarified in the title, introduction and discussion that results generated here may not be applied to rainfed systems, and that our focus was irrigated production systems in the Arkansas Delta region. The new text reads:

Title

"Synergistic maturity group and planting date adaptation sustains irrigated soybean yields under warming"

Introduction

"A comprehensive simulation study across 4,651 fields and 40 weather-years was conducted using a well-tested version of the APSIM (Agricultural Production Systems Simulator) framework. Irrigated soybean production systems in the Arkansas Delta region were considered a case study, an environment particularly vulnerable to warming (Sharma et al., 2023), with hot summers and frequent late-season drought events, which accounts for over 1.5 million hectares of soybean-planted area (NASS, 2025). The study's hypothesis was that under high temperatures, using late-maturing soybean genotypes combined with early planting will offset the negative impacts of warming on irrigated soybean yield."

Discussion

"Because this study focuses on irrigated soybean production systems in the Arkansas Delta region, the results generated may not be directly applicable to rainfed systems."

Comment #17

Along the same lines, the evaluated adaptation strategies assume an adequate irrigation water supply. However, the long-term availability of irrigation water may become a limitation in the Arkansas Delta due to increasing pressure on groundwater resources. The authors should briefly discuss this limitation or acknowledge that future work should evaluate adaptation strategies under limited irrigation or water-constrained scenarios.

Response: Thank you for the great suggestion. We have added a sentence to the discussion acknowledging this limitation and suggesting work to address this. The new text reads:

"Lastly, our simulations assume an adequate supply of irrigation water. Increasing pressure on groundwater resources raises concerns about the long-term availability of irrigation water in the study region. Future studies could focus on adaptive strategies under deficit-irrigation scenarios and on maximizing systems' water use efficiency."

Comment #18

The authors state that "extending the seed filling period was a promising strategy to maximize soybean seed yield" (lines 183–184). However, it is not clear how the seed-filling period was determined from APSIM. Please describe this in the Materials and Methods, including which phenological stages were used to calculate this duration.

Response: Thank you for the suggestion. We have clarified in the material and methods how seed-filling is determined in APSIM and added information on which stages were used to calculate seed-filling. The new sentences read:

In section "2.1. APSIM-Soybean model testing"

"The APSIM-Soybean model uses a multiplicative function to describe the daily relative development rate as a function of temperature and photoperiod (Archontoulis et al., 2014). The timing of key development phases is determined by integrating the daily development over time."

In section "2.2. Regional simulations and scenario analysis"

"In addition to seed yield (0% moisture), changes in the total crop cycle and seed-filling period were quantified. The seed filling period was the number of days from the beginning of seed (R5) to physiological maturity (R7), while the total crop cycle was the number of days from emergence (VE) to R7."

Comment #19

Were other developmental phases also affected by the adaptation strategies? Presenting the duration of the vegetative and reproductive phases (or their relative changes) would provide a clearer mechanistic explanation of the observed yield responses. As currently presented, it is not evident why extending the seed-filling period was the primary driver of increased yield.

Response: Thank you for the comment. In the discussion section, we stated that "extending the seed filling period was a promising strategy to maximize soybean seed yield ". We agree that the present analysis does not imply that the seed-filling period was the "primary" driver of the increase in yield. All the text was revised to ensure that no misleading conclusions were made.

To address your suggestion, we have created a new supplementary figure showing changes in duration of different phases, which complements Figure 2 of the manuscript. The new supplementary figure is shown below:

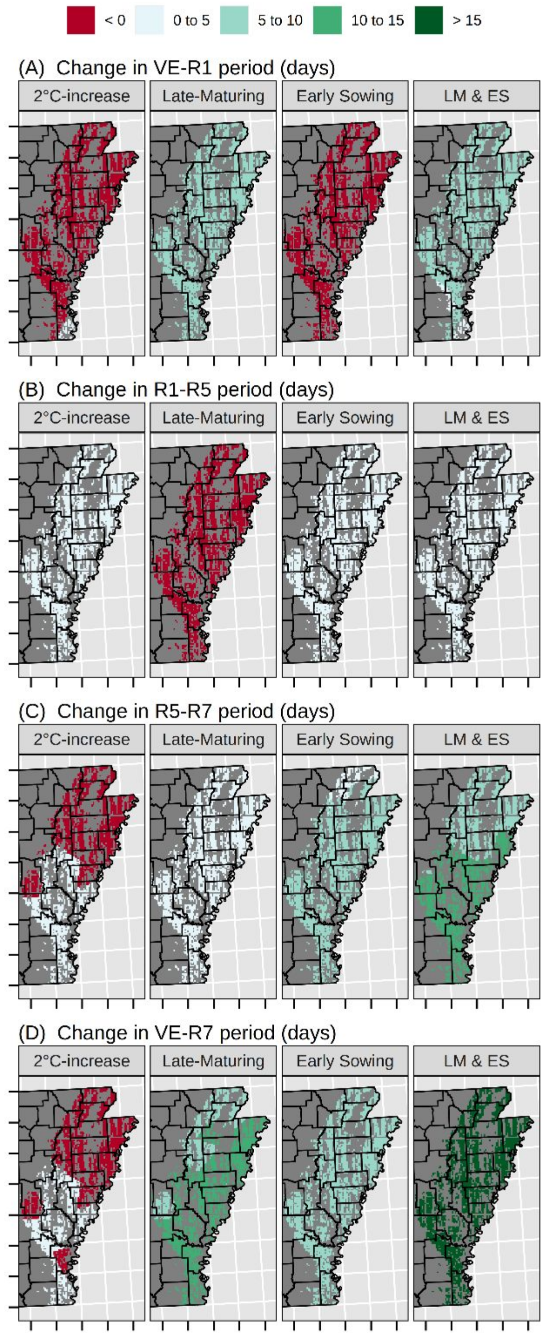

Figure S6. Geographical distribution of the relative effects of rising temperatures and adaptive genotypic and agronomic management strategies on changes in different soybean phenological phases in relation to the baseline: (A) emergence to first flower (VE-R1), (B) first flower to beginning seed (R1-R5), (C) beginning seed to physiological maturity (R5-R7), and (D) emergence to physiological maturity (VE-R7) across 4,651 soybean fields in the US Mid-South. Values per pixel are averages across 40 weather years. LM & ES represent the combined effect of a late-maturing variety and early sowing. All adaptive scenarios were simulated under +2°C.

Comparing phenological changes in the new figure with yield changes (Figure 1 in the manuscript) across scenarios, the additional analysis reinforces the message that the seed-filling period was associated with increased yields. However, further research is needed to determine whether the seed filling period is the primary driver of increased yield, a question beyond the scope of the present study.

Comment #20

In addition, please verify the figure citation in lines 183–184. It appears that Figure 2, rather than Figure 1, should be referenced.

Response: Thank you for catching this. Addressed.

SYNERGISTIC MATURITY GROUP AND PLANTING DATE ADAPTATION SUSTAINS IRRIGATED SOYBEAN YIELDS UNDER WARMING

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ABSTRACT

High temperatures increasingly threaten U.S. soybean seed yields, yet actionable solutions remain limited. T~~We evaluated t~~he APSIM model was evaluated against observed regional data from the U.S. Mid-South, and gridded simulations were conducted across 4,651 fields to identify how concurrent changes in planting date and maturity group affected yield under rising temperatures and elevated [CO_2]. The +2°C temperature scenario reduced yields by 5.7% on average. Combining a later-maturing soybean genotype with earlier sowing synergistically increased soybean yield by 12% under the +2°C scenario. These results revealed an opportunity to sustain soybean yield through management and genotype selection under warming. Advancing our understanding of management and genotypic effects on soybean phenology and improving their representation in process-based models are therefore essential for developing adaptive strategies that can support durable yield gains under high-temperature conditions.

1 INTRODUCTION

Rising temperatures and changes in rainfall patterns can decrease U.S. soybean seed yields by 19% in the absence of adaptive strategies (Elli et al., 2022). Higher temperatures accelerate phenological development, shorten~~ing~~ the crop cycle and limit~~ing~~ biomass accumulation and yield (Rose et al., 2016). Previous simulation studies indicate that adjusting soybean phenology may help offset yield reductions under global warming (Yang et al., 2025; Zabel et al., 2021). While projected increases in CO_2 concentration ([CO_2~~2~~]) can partially offset heat-induced yield losses (Ainsworth et al., 2002a; Kothari et al., 2022), complementary agronomic adaptation may be needed in some environments to fully offset the yield penalty (Makowski et al., 2020).

A relatively simple and well-established strategy to adjust the soybean life cycle to the available growing season length is to select soybean maturity groups (MGs) based on local temperature and photoperiod conditions (Mourtzinis and Conley, 2017). Moreover, earlier planting extends the growing season and shifts the crop cycle into cooler periods, reducing exposure to late-season drought and heat events (Edwards et al., 2003; Bastidas et al., 2008). While farmers have been shifting towards earlier planting and maturity groups well-adapted to a region, the synergistic effect of these shifts under a warming scenario remains unquantified.

~~We conducted a~~A comprehensive simulation study across 4,651 fields and 40 weather-years was conducted using a well-tested version of the APSIM (Agricultural Production Systems Simulator) framework. ~~We considered~~ Irrigated soybean production systems in the Arkansas Delta region were considered a case study, an environment particularly vulnerable to warming (Sharma et al., 2023), with hot summers and frequent late-season drought events, which accounts for over 1.5 million hectares of soybean-planted area (NASS, 2025). ~~We~~ The study's hypothesis was~~ze~~ that under high temperatures, using late-maturing soybean genotypes combined with early planting will offset the negative impacts of warming on irrigated soybean yield. Our objective was to quantify the individual and synergistic impacts of changes in maturity group and sowing date on adapting soybean production to high-temperature risks under current and elevated [CO_2].

2 MATERIAL AND METHODS

2.1. APSIM-Soybean model testing

The soybean model within APSIM Next Generation (Holzworth et al., 2018) was evaluated against regional data from the U.S. Midsouth covering multiple locations, planting dates, and cultivar maturity groups (Figure S1), as described in Salmerón et al. (2016). The APSIM-Soybean model uses a multiplicative function to describe the daily relative development rate as a function of temperature and photoperiod (Archontoulis et al., 2014). The timing of key development phases is determined by integrating the daily development over time. The ~~APSIM-Soybean~~ model ~~has~~ was previously ~~proven~~ successful in simulating the growth of various soybean maturity groups under different environmental conditions in the US (Archontoulis et al., 2020; Elli et al., 2022). We further tested the model against experimental data and updated cultivar coefficients to the latest APSIM Next Generation (Holzworth et al., 2018), version 2025.3.7681. Cultivar coefficients were calibrated for an early MG 4 and an early MG 5 cultivar. The calibration results confirmed that the model was robust across a wide range of environmental conditions (Figures S1 and S2 and Table S1), with RMSE values similar to those reported in other studies on soybean phenology and yield (Archontoulis et al., 2014; Salmerón and Purcell, 2016).

2.2. Regional simulations and scenario analysis

~~We conducted a~~A simulation exercise across 40 weather years and 4,651 fields in the US Mid-South was conducted. Further information on the weather conditions of the study area is presented in Figure S3. T~~We used t~~he soybean cropland data layer (NASS, 2025) was used to identify soybean-cropped pixels and define the spatial extent of the simulation fields, which were resampled to a 2.5 km grid (~0.025°). Each grid cell classified as soybean cropland was treated as one simulation field. We used the *apsimx* R package (Miguez, 2020) to run the simulations. Daily historical weather data on maximum and minimum temperatures, solar radiation, and rainfall were retrieved from the nearest weather station in the National Weather Service Network using the *apsimx* R package. [CO2] was set at 350 ppm. Soil profile information was derived from SSURGO (Soil Survey Staff, 2023). Within each simulated grid cell, we used the most predominant soil profile. Soil water content was reset at field capacity on January 1 each year to avoid confounding carryover effects (Elli and Archontoulis, 2023). For baseline simulations, the sowing date was May 22, an average value for the studied region based on USDA-NASS 50% sowing progress from 1990 to 2025 (NASS, 2025; Figure S4). We used an early MG 4 cultivar, which is well adapted to the region (Salmerón et al., 2016; Salmerón et al., 2014). Automatic irrigation was triggered between R1 and R7 when soil water reached 50% of the plant-available water in the 0-60 cm depth, to refill soil moisture to field capacity.

For the high-temperature scenario, ~~we considered a~~ 2°C ~~increase (+2°C) in~~was added to the daily maximum and minimum temperatures from the historical weather station data.~~, which is~~This +2°C value represents an average projection derived from 15 global circulation models for 2050, as reported by Elli et al. (2022). A +2°C scenario combined with an elevated [CO2] to 540 ppm (Elli and Archontoulis, 2023) was also evaluated. The +2°C scenario ~~We employed~~is a simpler approach compared to ~~, rather than~~using ~~the~~ direct outputs of global circulation climate models, but ~~because~~ it has ~~been shown to yield~~produced results with similar direction and magnitude while substantially reducing computational requirements (Elli et al., 2022). The adaptive scenarios included (1) an early MG 5 cultivar instead of early MG 4, (2) an early sowing date by 4 weeks (April 24), corresponding to the average USDA-NASS 10% sowing progress from 1990 to 2025 (NASS, 2025; Figure S4), and (3) a combination of an early MG 5 cultivar with early sowing. Data analysis and visualization were conducted using R version 4.1.1 (R Development Core Team, 2024). In addition to seed yield (0% moisture), changes in the total crop cycle and seed-filling period were quantified. The seed filling period was the number of days from the beginning of seed (R5) to physiological maturity (R7), while the total crop cycle was the number of days from emergence (VE) to R7.

3 RESULTS

3.1. Elevated temperature and [CO2] impacts on soybean yields without adaptation

The +2°C temperature scenario decreased soybean seed yields by 5.7% on average (~245 kg ha^{-1}), with values ranging from -14.7% to -2.5% across the 4,651 ~~sites~~ fields (Figure 1). The mean simulated yield for the baseline condition was 4,300 kg ha^{-1} with a south-to-north geographical gradient ranging from 2,142 to 4,837 kg ha^{-1} (Figure S5). Elevated [CO_2] consistently offset the negative effects of higher temperatures across ~~sites~~ fields (Figure 1), with an average yield 12.9% greater than under baseline conditions, and values ranging from 9.2% to 15.4% across all ~~sites~~ fields (Figure 1). A shortening of the total crop cycle was observed in 61% of the sites, with more pronounced changes in northern environments (Figure 2), where baseline temperatures ~~are~~ were lower. The total crop cycle length decreased by up to 15 days across all ~~site~~field-years. The seed filling period was shortened by up to 9 days across all ~~site~~field-years.

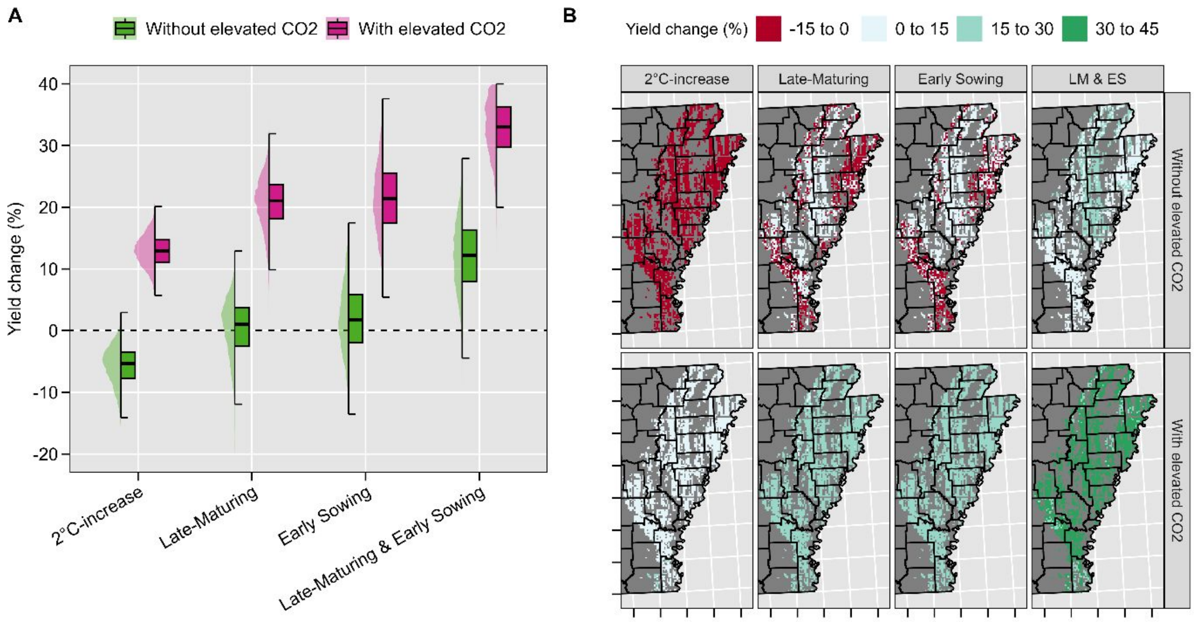

Figure 1. Overall relative effects of rising temperatures and adaptive genotypic and agronomic management strategies on soybean seed yield across 4,651 soybean fields and 40 weather years in the US Mid-South (A), and their geographical distribution (B). Values per pixel in panel B are averages across 40 weather years. LM & ES represent the combined effect of a late-maturing variety and early sowing. All adaptive scenarios were simulated under +2°C.

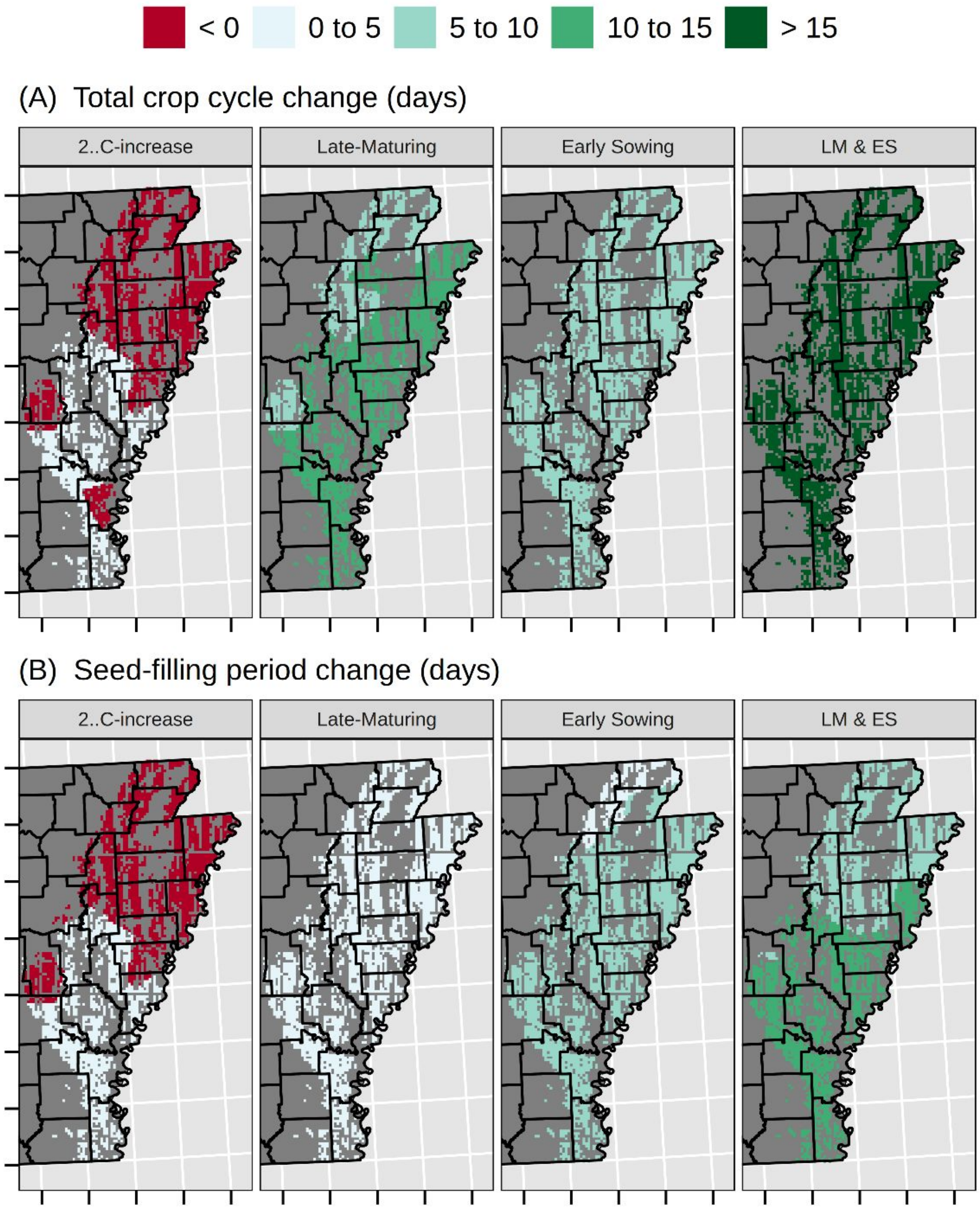

Figure 2. Geographical distribution of the relative effects of rising temperatures and adaptive genotypic and agronomic management strategies on changes in soybean total crop cycle (A) and seed-filling period (B) compared to baseline across 4,651 soybean fields in the US Mid-South. Values per pixel are averages across 40 weather years. LM & ES represent the combined effect of a late-maturing variety and early sowing. All adaptive scenarios were simulated under +2°C.

3.2. Adaptive strategies to elevated temperature

Stacked agronomic management strategies were more effective than standalone strategies to maximize soybean yields under warming (Figure 1). Using a late-maturing soybean cultivar in combination with early sowing increased soybean seed yield by 12% under the +2°C temperature scenario, without elevated CO_2. The combined adaptive scenario extended the seed-filling period by an average of 10.4 days (Figure 2). The standalone strategy of using a late-maturing cultivar under a +2°C temperature scenario resulted in average yields comparable to the historical baseline, with values ranging from -12.6% to 4.8% across fields. This practice increased yields at 62% of the 4,651 simulated ~~sites~~fields, particularly in the central-north region (Figure 1B). The Early sowing strategy increased yields by an average of 2% relative to the baseline, with values ranging from -4.9% to 7.8%. This practice increased yields at 71% of the ~~sites~~fields. Elevated [CO_2] altered the magnitude but not the direction of the yield response to stacked adaptive management options. The combined impact of elevated CO_2 and stacked agronomic management strategies increased yields by an average of 34%, with values ranging from 26.1% to 41.8% (Figure 1).

4 DISCUSSION

Our study revealed that soybean yields under warming can be sustained by concurrently adjusting cultivar maturity and planting date. Under the +2°C scenario, the standalone strategies of using a late-maturing cultivar and planting earlier increased yield by an average of 0.5% and 2%, respectively, relative to the historical simulations. This implies that individual strategies offset the impacts of higher temperature, resulting in yields comparable to baseline yields under current weather conditions. However, the offset was not consistent across all ~~sites~~ fields (Figure 1). The combined strategies increased seed yield by an average of 12% in relation to the baseline, with values ranging from 0.4% to 17.6% across ~~sites~~fields, demonstrating a synergistic impact and emerging as the most promising approach to sustain soybean yield under a +2°C scenario. Our study extends previous work (Elli et al., 2022) by quantifying the synergistic effect of stacked agronomic management strategies to mitigate heat risks in irrigated soybeans at a fine spatial resolution in the U.S. Mid-South.

~~We found a~~A 5.7% decrease in soybean yield was found due to rising temperatures, with values ranging from -14.7% to -2.5% across ~~sites~~ fields (Figure 1). Elli et al. (2022) reported an average 19% reduction in seed yield at 10 locations in the US Corn Belt under a +2°C temperature scenario without adaptive strategies. While Elli et al. (2022) did not quantify the combined impact of high temperatures and stacked agronomic management strategies, they found that sowing 12 days earlier increased soybean seed yield by 4.7%. Deryng et al. (2014) investigated the effects of heat stress on soybeans worldwide and predicted a 33% yield reduction by 2080.

While elevated [CO_2] can offset the negative impacts of rising temperatures (Ainsworth et al., 2002b), structural model uncertainties exist in predicting yield benefits under elevated [CO_2] (Kothari et al., 2022). Recognizing that soybeans are already a highly heat-tolerant crop, heat events above 28°C during the reproductive phase can reduce yield by lowering seed number and weight (Boote et al., 2005; Djanaguiraman et al., 2013). The APSIM-soybean version used in the present study accounted~~s~~ for temperature effects on crop growth rate (Kothari et al., 2024), but did~~oes~~ not account for the impacts of extreme heat on seed number determination, a common limitation in process-based models (Richetti et al., 2025). Moreover, APSIM assumes constant temperature responses throughout the life cycle, whereas in soybeans this response may change across developmental stages (Hatfield et al., 2011).

Warmer springs and longer frost-free periods have allowed farmers to plant earlier every year (Mourtzinis et al., 2019). However, this practice is not always feasible due to extreme rainfall events, which are projected to intensify further (Elli et al., 2022). Extending the seed filling period was promising to maximize soybean seed yield (Figures 2 and S6~~1~~), which agrees with the findings of Boote et al. (2003). However, phenology did not fully explain yield reduction under warming. This could be associated with slower development on days when temperatures exceed the optimum temperature for soybean development in APSIM (30°C, Figure S3). Battisti et al. (2018) observed that, owing to the temperature response function governing phenological development, the APSIM model simulated a greater reduction in soybean cycle length under a +4.5°C warming scenario than under a +6.0°C scenario in Southern Brazil, with a mean air temperature 23.65°C. Further yield decreases could be associated with high-temperature effects on radiation use efficiency (Kothari et al., 2022; Ruiz-Vera et al., 2013).

While exploring optimum planting date windows for target environments was not the focus of this study, future research could investigate more extreme scenarios to enhance mechanistic understanding of yield responses under warming. Because this study focuses on irrigated soybean production systems in the Arkansas Delta region, the results generated may not be directly applicable to rainfed systems. Lastly, our simulations assume an adequate supply of irrigation water. Increasing pressure on groundwater resources raises concerns about the long-term availability of irrigation water in the study region. Future studies could focus on adaptive strategies under deficit-irrigation scenarios and on maximizing systems' water use efficiency.

5 CONCLUSION

This study advances our understanding of synergistic agronomic management adaptation to sustain soybean seed yields under warming, providing new quantitative evidence from ~~the~~ irrigated s~~S~~oybean~~s~~ in the US Mid-South that extends the existing literature. ~~We found that~~ ~~s~~Soybean yields declined by 5.7% on average under rising temperatures. Adjusting the soybean maturity group, combined with early sowing, resulted in an overall yield gain of 12% relative to the historical baseline.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

AUTHOR CONTRIBUTIONS

Elvis F. Elli: Conceptualization, Data curation, Formal analysis, Funding acquisition, Visualization, Writing - original draft; Caio dos Santos: Formal analysis, Methodology, Software Writing - review & editing; Samuel B. Fernandes: Methodology, Software, Writing - review & editing; Montserrat Salmerón: Investigation, Methodology, Resources, Writing - review & editing; Rafael Battisti: Methodology, Writing - review & editing; Rogério de S. Nóia-Júnior: Conceptualization, Investigation, Writing - review & editing.

ACKNOWLEDGMENTS

This study was supported by the University of Arkansas System Division of Agriculture (Elli) and FFAR (Fernandes, project title: Integrating genomic prediction with crop modeling to develop climate-resilient cultivars). Sincere appreciation is extended to all researchers who collected the phenology data used for model testing.

SUPPLEMENTAL MATERIAL

Supplemental materials are included in this paper.

Figure S1. Locations of multi-environment trials used to collect phenology and yield data for soybean maturity groups 4 to 5 (A). Relationship between APSIM-simulated and observed day of year (DOY) for key phenological stages: emergence, beginning of flowering, beginning of pod development, beginning of seed filling, and physiological maturity (B). Relationship between APSIM-simulated and observed soybean yield at 13% grain moisture (C).

Figure S2. APSIM-simulated (lines) and measured (points) leaf biomass (green colour), seed biomass (black colour), and total biomass (yellow colour) in Fayetteville, Northwest Arkansas, for 2 different maturity groups (early 4 and early 5). Measured data were obtained from an experiment conducted in 2008 under irrigated conditions in a randomized complete block design with 4 replications. Further information on field data collection can be found in (Mastrodomenico and Purcell, 2012). Vertical lines represent the standard deviations. RMSEs for seed yield, leaf biomass, and total biomass were 659 kg/ha, 572 kg/ha, and 1335 kg/ha, respectively.

Figure S3. Environmental characterization of baseline conditions across all studied ~~site~~field-years. The APSIM phenology response curve (0-1) is presented, where 1 represents maximum development rate. The solid vertical and horizontal lines indicate the median growing-season average temperature and total growing-season precipitation, respectively.

Figure S4. USDA-NASS soybean sowing progress in Arkansas progress from 1990 to 2025. Grey lines represent the historical years, while the green, thicker line indicates the average sowing progress. 50% and 10% average sowing progress are indicated in the figure.

Figure S5. Density distribution of simulated soybean yields without adaptation under the baseline and +2°C scenarios with and without elevated [CO2] (A), and its geographical distribution (B).

Figure S6. Geographical distribution of the relative effects of rising temperatures and adaptive genotypic and agronomic management strategies on changes in different soybean phenological phases in relation to the baseline: (A) emergence to first flower (VE-R1), (B) first flower to beginning seed (R1-R5), (C) beginning seed to physiological maturity (R5-R7), and (D) emergence to physiological maturity (VE-R7) across 4,651 soybean fields in the US Mid-South. Values per pixel are averages across 40 weather years. LM & ES represent the combined effect of a late-maturing variety and early sowing. All adaptive scenarios were simulated under +2°C.

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SYNERGISTIC MATURITY GROUP AND PLANTING DATE ADAPTATION SUSTAINS IRRIGATED SOYBEAN YIELDS UNDER WARMING

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ABSTRACT

High temperatures increasingly threaten U.S. soybean seed yields, yet actionable solutions remain limited. The APSIM model was evaluated against observed regional data from the U.S. Mid-South, and gridded simulations were conducted across 4,651 fields to identify how concurrent changes in planting date and maturity group affected yield under rising temperatures and elevated [CO_2]. The +2°C temperature scenario reduced yields by 5.7% on average. Combining a later-maturing soybean genotype with earlier sowing synergistically increased soybean yield by 12% under the +2°C scenario. These results revealed an opportunity to sustain soybean yield through management and genotype selection under warming. Advancing our understanding of management and genotypic effects on soybean phenology and improving their representation in process-based models are therefore essential for developing adaptive strategies that can support durable yield gains under high-temperature conditions.

1 INTRODUCTION

Rising temperatures and changes in rainfall patterns can decrease U.S. soybean seed yields by 19% in the absence of adaptive strategies (Elli et al., 2022). Higher temperatures accelerate phenological development, shorten the crop cycle and limit biomass accumulation and yield (Rose et al., 2016). Previous simulation studies indicate that adjusting soybean phenology may help offset yield reductions under global warming (Yang et al., 2025; Zabel et al., 2021). While projected increases in CO_2 concentration ([CO_2]) can partially offset heat-induced yield losses (Ainsworth et al., 2002a; Kothari et al., 2022), complementary agronomic adaptation may be needed in some environments to fully offset the yield penalty (Makowski et al., 2020).

A relatively simple and well-established strategy to adjust the soybean life cycle to the available growing season length is to select soybean maturity groups (MGs) based on local temperature and photoperiod conditions (Mourtzinis and Conley, 2017). Moreover, earlier planting extends the growing season and shifts the crop cycle into cooler periods, reducing exposure to late-season drought and heat events (Edwards et al., 2003; Bastidas et al., 2008). While farmers have been shifting towards earlier planting and maturity groups well-adapted to a region, the synergistic effect of these shifts under a warming scenario remains unquantified.

A comprehensive simulation study across 4,651 fields and 40 weather-years was conducted using a well-tested version of the APSIM (Agricultural Production Systems Simulator) framework. Irrigated soybean production systems in the Arkansas Delta region were considered a case study, an environment particularly vulnerable to warming (Sharma et al., 2023), with hot summers and frequent late-season drought events, which accounts for over 1.5 million hectares of soybean-planted area (NASS, 2025). The study's hypothesis was that under high temperatures, using late-maturing soybean genotypes combined with early planting will offset the negative impacts of warming on irrigated soybean yield. Our objective was to quantify the individual and synergistic impacts of changes in maturity group and sowing date on adapting soybean production to high-temperature risks under current and elevated [CO_2].

2 MATERIAL AND METHODS

2.1. APSIM-Soybean model testing

The soybean model within APSIM Next Generation (Holzworth et al., 2018) was evaluated against regional data from the U.S. Midsouth covering multiple locations, planting dates, and cultivar maturity groups (Figure S1), as described in Salmerón et al. (2016). The APSIM-Soybean model uses a multiplicative function to describe the daily relative development rate as a function of temperature and photoperiod (Archontoulis et al., 2014). The timing of key development phases is determined by integrating the daily development over time. The model was previously successful in simulating the growth of various soybean maturity groups under different environmental conditions in the US (Archontoulis et al., 2020; Elli et al., 2022). We further tested the model against experimental data and updated cultivar coefficients to the latest APSIM Next Generation (Holzworth et al., 2018), version 2025.3.7681. Cultivar coefficients were calibrated for an early MG 4 and an early MG 5 cultivar. The calibration results confirmed that the model was robust across a wide range of environmental conditions (Figures S1 and S2 and Table S1), with RMSE values similar to those reported in other studies on soybean phenology and yield (Archontoulis et al., 2014; Salmerón and Purcell, 2016).

2.2. Regional simulations and scenario analysis

A simulation exercise across 40 weather years and 4,651 fields in the US Mid-South was conducted. Further information on the weather conditions of the study area is presented in Figure

S3. The soybean cropland data layer (NASS, 2025) was used to identify soybean-cropped pixels and define the spatial extent of the simulation fields, which were resampled to a 2.5 km grid (~0.025°). Each grid cell classified as soybean cropland was treated as one simulation field. We used the *apsimx* R package (Miguez, 2020) to run the simulations. Daily historical weather data on maximum and minimum temperatures, solar radiation, and rainfall were retrieved from the nearest weather station in the National Weather Service Network using the *apsimx* R package. [CO2] was set at 350 ppm. Soil profile information was derived from SSURGO (Soil Survey Staff, 2023). Within each simulated grid cell, we used the most predominant soil profile. Soil water content was reset at field capacity on January 1 each year to avoid confounding carryover effects (Elli and Archontoulis, 2023). For baseline simulations, the sowing date was May 22, an average value for the studied region based on USDA-NASS 50% sowing progress from 1990 to 2025 (NASS, 2025; Figure S4). We used an early MG 4 cultivar, which is well adapted to the region (Salmerón et al., 2016; Salmerón et al., 2014). Automatic irrigation was triggered between R1 and R7 when soil water reached 50% of the plant-available water in the 0-60 cm depth, to refill soil moisture to field capacity.

For the high-temperature scenario, 2°C was added to the daily maximum and minimum temperatures from the historical weather station data. This +2°C value represents an average projection derived from 15 global circulation models for 2050, as reported by Elli et al. (2022). A +2°C scenario combined with an elevated [CO2] to 540 ppm (Elli and Archontoulis, 2023) was also evaluated. The +2°C scenario is a simpler approach compared to using direct outputs of global circulation climate models, but it has produced results with similar direction and magnitude while substantially reducing computational requirements (Elli et al., 2022). The adaptive scenarios included (1) an early MG 5 cultivar instead of early MG 4, (2) an early sowing date by 4 weeks (April 24), corresponding to the average USDA-NASS 10% sowing progress from 1990 to 2025 (NASS, 2025; Figure S4), and (3) a combination of an early MG 5 cultivar with early sowing. Data analysis and visualization were conducted using R version 4.1.1 (R Development Core Team, 2024). In addition to seed yield (0% moisture), changes in the total crop cycle and seed-filling period were quantified. The seed filling period was the number of days from the beginning of seed (R5) to physiological maturity (R7), while the total crop cycle was the number of days from emergence (VE) to R7.

3 RESULTS

3.1. Elevated temperature and [CO2] impacts on soybean yields without adaptation

The +2°C temperature scenario decreased soybean seed yields by 5.7% on average (~245 kg ha^{-1}), with values ranging from -14.7% to -2.5% across the 4,651 fields (Figure 1). The mean simulated yield for the baseline condition was 4,300 kg ha^{-1} with a south-to-north geographical gradient ranging from 2,142 to 4,837 kg ha^{-1} (Figure S5). Elevated [CO_2] consistently offset the negative effects of higher temperatures across fields (Figure 1), with an average yield 12.9% greater than under baseline conditions, and values ranging from 9.2% to 15.4% across all fields (Figure 1). A shortening of the total crop cycle was observed in 61% of the sites, with more pronounced changes in northern environments (Figure 2), where baseline temperatures were lower. The total crop cycle length decreased by up to 15 days across all field-years. The seed filling period was shortened by up to 9 days across all field-years.

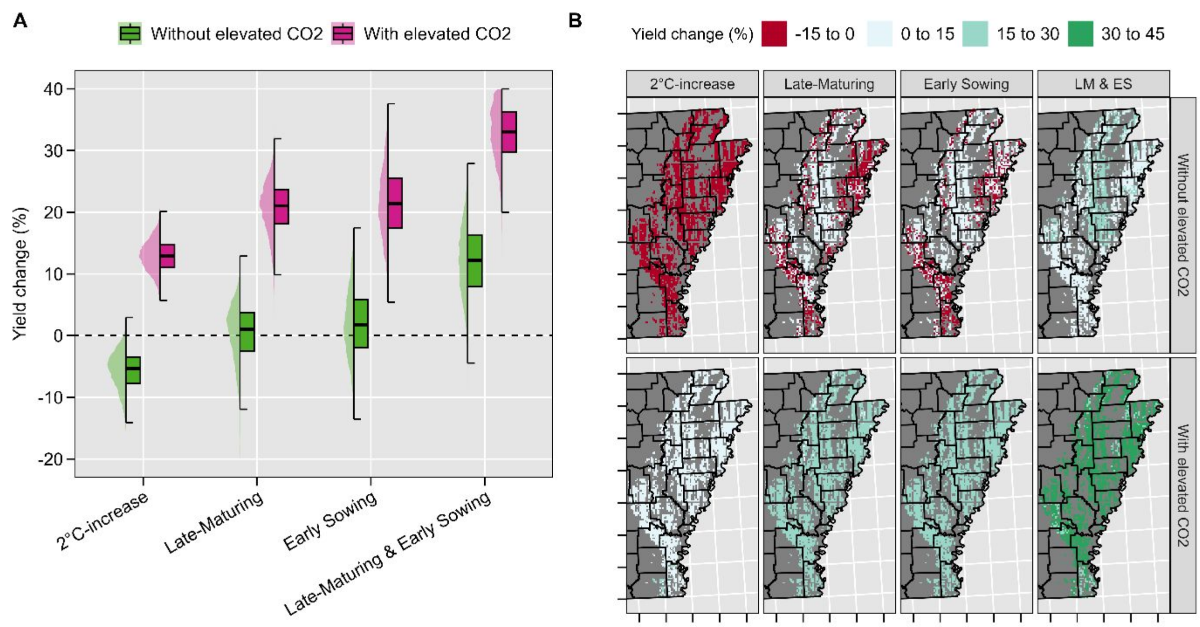

Figure 1. Overall relative effects of rising temperatures and adaptive genotypic and agronomic management strategies on soybean seed yield across 4,651 soybean fields and 40 weather years in the US Mid-South (A), and their geographical distribution (B). Values per pixel in panel B are averages across 40 weather years. LM & ES represent the combined effect of a late-maturing variety and early sowing. All adaptive scenarios were simulated under +2°C.

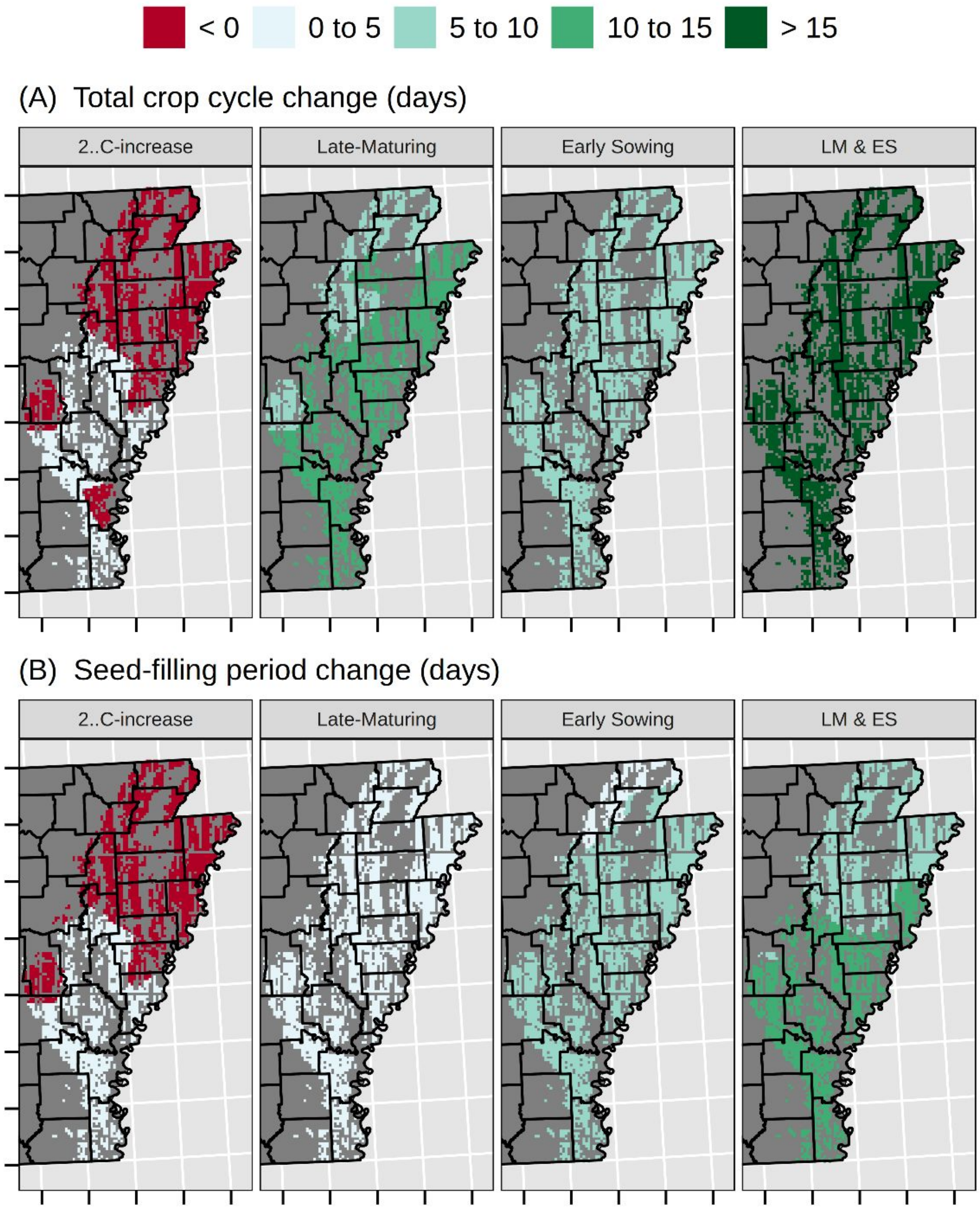

Figure 2. Geographical distribution of the relative effects of rising temperatures and adaptive genotypic and agronomic management strategies on changes in soybean total crop cycle (A) and seed-filling period (B) compared to baseline across 4,651 soybean fields in the US Mid-South. Values per pixel are averages across 40 weather years. LM & ES represent the combined effect of a late-maturing variety and early sowing. All adaptive scenarios were simulated under +2°C.

3.2. Adaptive strategies to elevated temperature

Stacked agronomic management strategies were more effective than standalone strategies to maximize soybean yields under warming (Figure 1). Using a late-maturing soybean cultivar in combination with early sowing increased soybean seed yield by 12% under the +2°C temperature scenario, without elevated CO_2. The combined adaptive scenario extended the seed-filling period by an average of 10.4 days (Figure 2). The standalone strategy of using a late-maturing cultivar under a +2°C temperature scenario resulted in average yields comparable to the historical baseline, with values ranging from -12.6% to 4.8% across fields. This practice increased yields at 62% of the 4,651 simulated fields, particularly in the central-north region (Figure 1B). The Early sowing strategy increased yields by an average of 2% relative to the baseline, with values ranging from -4.9% to 7.8%. This practice increased yields at 71% of the fields. Elevated [CO_2] altered the magnitude but not the direction of the yield response to stacked adaptive management options. The combined impact of elevated CO_2 and stacked agronomic management strategies increased yields by an average of 34%, with values ranging from 26.1% to 41.8% (Figure 1).

4 DISCUSSION

Our study revealed that soybean yields under warming can be sustained by concurrently adjusting cultivar maturity and planting date. Under the +2°C scenario, the standalone strategies of using a late-maturing cultivar and planting earlier increased yield by an average of 0.5% and 2%, respectively, relative to the historical simulations. This implies that individual strategies offset the impacts of higher temperature, resulting in yields comparable to baseline yields under current weather conditions. However, the offset was not consistent across all fields (Figure 1). The combined strategies increased seed yield by an average of 12% in relation to the baseline, with values ranging from 0.4% to 17.6% across fields, demonstrating a synergistic impact and emerging as the most promising approach to sustain soybean yield under a +2°C scenario. Our study extends previous work (Elli et al., 2022) by quantifying the synergistic effect of stacked agronomic management strategies to mitigate heat risks in irrigated soybeans at a fine spatial resolution in the U.S. Mid-South.

A 5.7% decrease in soybean yield was found due to rising temperatures, with values ranging from -14.7% to -2.5% across fields (Figure 1). Elli et al. (2022) reported an average 19% reduction in seed yield at 10 locations in the US Corn Belt under a +2°C temperature scenario without adaptive strategies. While Elli et al. (2022) did not quantify the combined impact of high temperatures and stacked agronomic management strategies, they found that sowing 12 days earlier increased soybean seed yield by 4.7%. Deryng et al. (2014) investigated the effects of heat stress on soybeans worldwide and predicted a 33% yield reduction by 2080.

While elevated [CO_2] can offset the negative impacts of rising temperatures (Ainsworth et al., 2002b), structural model uncertainties exist in predicting yield benefits under elevated [CO_2] (Kothari et al., 2022). Recognizing that soybeans are already a highly heat-tolerant crop, heat events above 28°C during the reproductive phase can reduce yield by lowering seed number and weight (Boote et al., 2005; Djanaguiraman et al., 2013). The APSIM-soybean version used in the present study accounted for temperature effects on crop growth rate (Kothari et al., 2024), but did not account for the impacts of extreme heat on seed number determination, a common limitation in process-based models (Richetti et al., 2025). Moreover, APSIM assumes constant temperature responses throughout the life cycle, whereas in soybeans this response may change across developmental stages (Hatfield et al., 2011).

Warmer springs and longer frost-free periods have allowed farmers to plant earlier every year (Mourtzinis et al., 2019). However, this practice is not always feasible due to extreme rainfall events, which are projected to intensify further (Elli et al., 2022). Extending the seed filling period was promising to maximize soybean seed yield (Figures 2 and S6), which agrees with the findings of Boote et al. (2003). However, phenology did not fully explain yield reduction under warming. This could be associated with slower development on days when temperatures exceed the optimum temperature for soybean development in APSIM (30°C, Figure S3). Battisti et al. (2018) observed that, owing to the temperature response function governing phenological development, the APSIM model simulated a greater reduction in soybean cycle length under a +4.5°C warming scenario than under a +6.0°C scenario in Southern Brazil, with a mean air temperature 23.65°C. Further yield decreases could be associated with high-temperature effects on radiation use efficiency (Kothari et al., 2022; Ruiz-Vera et al., 2013).

While exploring optimum planting date windows for target environments was not the focus of this study, future research could investigate more extreme scenarios to enhance mechanistic understanding of yield responses under warming. Because this study focuses on irrigated soybean production systems in the Arkansas Delta region, the results generated may not be directly applicable to rainfed systems. Lastly, our simulations assume an adequate supply of irrigation water. Increasing pressure on groundwater resources raises concerns about the long-term availability of irrigation water in the study region. Future studies could focus on adaptive strategies under deficit-irrigation scenarios and on maximizing systems' water use efficiency.

5 CONCLUSION

This study advances our understanding of synergistic agronomic management adaptation to sustain soybean seed yields under warming, providing new quantitative evidence from irrigated soybean in the US Mid-South that extends the existing literature. Soybean yields declined by

5.7% on average under rising temperatures. Adjusting the soybean maturity group, combined with early sowing, resulted in an overall yield gain of 12% relative to the historical baseline.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

AUTHOR CONTRIBUTIONS

Elvis F. Elli: Conceptualization, Data curation, Formal analysis, Funding acquisition, Visualization, Writing - original draft; Caio dos Santos: Formal analysis, Methodology, Software Writing - review & editing; Samuel B. Fernandes: Methodology, Software, Writing - review & editing; Montserrat Salmerón: Investigation, Methodology, Resources, Writing - review & editing; Rafael Battisti: Methodology, Writing - review & editing; Rogério de S. Nóia-Júnior: Conceptualization, Investigation, Writing - review & editing.

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SUPPLEMENTAL MATERIAL

Supplemental materials are included in this paper.

Figure S1. Locations of multi-environment trials used to collect phenology and yield data for soybean maturity groups 4 to 5 (A). Relationship between APSIM-simulated and observed day of year (DOY) for key phenological stages: emergence, beginning of flowering, beginning of pod development, beginning of seed filling, and physiological maturity (B). Relationship between APSIM-simulated and observed soybean yield at 13% grain moisture (C).

Figure S2. APSIM-simulated (lines) and measured (points) leaf biomass (green colour), seed biomass (black colour), and total biomass (yellow colour) in Fayetteville, Northwest Arkansas, for 2 different maturity groups (early 4 and early 5). Measured data were obtained from an experiment conducted in 2008 under irrigated conditions in a randomized complete block design with 4 replications. Further information on field data collection can be found in (Mastrodomenico and Purcell, 2012). Vertical lines represent the standard deviations. RMSEs for seed yield, leaf biomass, and total biomass were 659 kg/ha, 572 kg/ha, and 1335 kg/ha, respectively.

Figure S3. Environmental characterization of baseline conditions across all studied field-years. The APSIM phenology response curve (0-1) is presented, where 1 represents maximum development rate. The solid vertical and horizontal lines indicate the median growing-season average temperature and total growing-season precipitation, respectively.

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Supplementary Material

SYNERGISTIC MATURITY GROUP AND PLANTING DATE ADAPTATION SUSTAINS IRRIGATED SOYBEAN YIELDS UNDER WARMING

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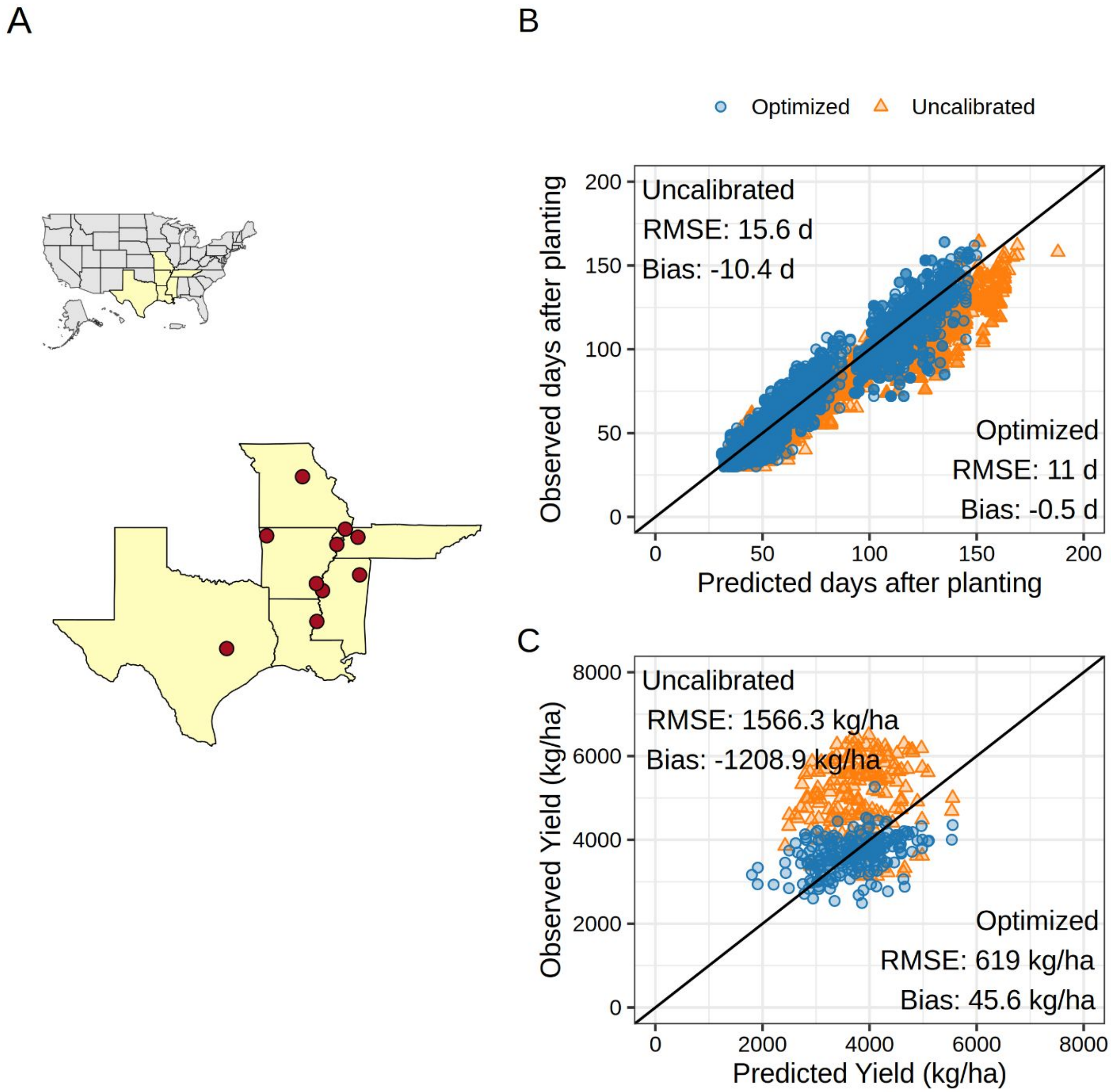


Figure S1. Locations of multi-environment trials used to collect phenology and yield data for soybean maturity groups 4 to 5 (A). Relationship between APSIM-simulated and observed days after sowing for key phenological stages: emergence, beginning of flowering, beginning of pod development, beginning of seed filling, and physiological maturity (B). Relationship between APSIM-simulated and observed soybean yield at 13% grain moisture (C).

Model calibration
The occurrence of phenological stages, converted to days after sowing, was used to derive the cultivar-specific model parameters (Table S1) using an iterative optimization procedure based on the Nelder–Mead algorithm (Nelder and Mead, 1965). This algorithm explores the parameter space iteratively using a geometric figure called simplex. The optimization procedure aims to minimize the residual sums of squares (Wallach et al., 2001) between the estimated and actual timing to reach the studied phenology stages for each maturity group studied (early 4, consisting of cultivars with relative MG 4.0 to 4.4, and an early MG 5, consisting of cultivars with relative MG 5.0 to 5.4). We ran the algorithm using R software (R Development Core Team, 2021) and the APSIMX package (Miguez, 2020). Our focus was on optimizing phenology parameters while keeping cardinal temperature-related parameters constant, giving little evidence of changes across genotypes (Archontoulis et al., 2014). Two growth parameters were selected: area of the largest leaf and radiation use efficiency. The optimization workflow followed an approach similar to that outlined in Wallach et al. (2019), in which all parameters were optimized simultaneously to potentially capture parameter interactions.

Table S1. Maturity group (MG) – specific default and optimized APSIM parameter values targeted in the optimization of maturity groups 4 and 5 cultivars.

APSIM Parameter	Units	Early MG 4	Early MG 5
Vegetative Target	°C-day	484	634
Early Flowering Target	°C-day	185	189
Early Grain Filling Target	Fraction (0-1)	0.338	0.042
Late Grain Filling Target	°C-day	384	418
Reproductive Photoperiod Modifier	Fraction (0-1)	11.48, 17.24	12.70, 14.31
Area of Largest Leaf	m^2	0.004	0.004
Radiation Use Efficiency	g MJ^{-1}	1.48	1.05

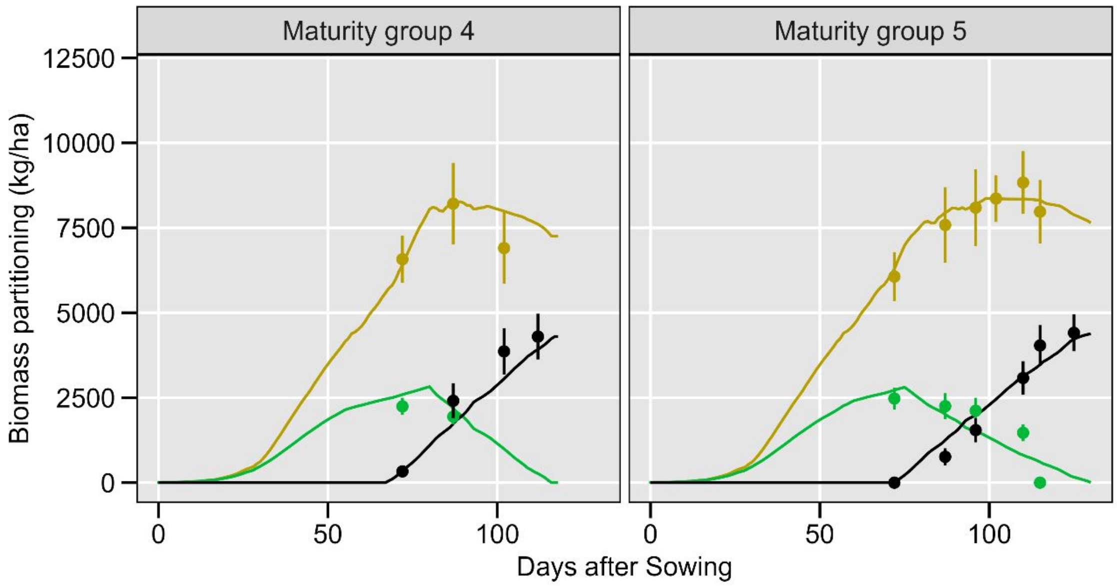


Figure S2. APSIM-simulated (lines) and measured (points) leaf biomass (green colour), seed biomass (black colour), and total biomass (yellow colour) in Fayetteville, Northwest Arkansas, for 2 different maturity groups (early 4 and early 5). Measured data were obtained from an experiment conducted in 2008 under irrigated conditions in a randomized complete block design with 4 replications. Further information on field data collection can be found in (Mastrodomenico and Purcell, 2012). Vertical lines represent the standard deviations. RMSEs for seed yield, leaf biomass, and total biomass were 659 kg/ha, 572 kg/ha, and 1335 kg/ha, respectively.

Study location and weather

The studied region is predominantly irrigated, with accumulated rainfall of 590 mm, unevenly distributed over the growing season (May to October). The average growing season temperature is 24°C, with values ranging from 20.9 to 27.1°C across all field-years (Figure S1). The studied area exhibits significant variability in soil conditions, ranging from shallow (less than 50 cm in depth) sandy soils with low soil water-holding capacity (0.4 mm cm^{-1}) to deeper soils with an effective root depth greater than 100 cm and high soil water-holding capacity (2 mm cm^{-1}). The soil organic carbon (0-30 cm) typically ranges from 0.5 to 3%.

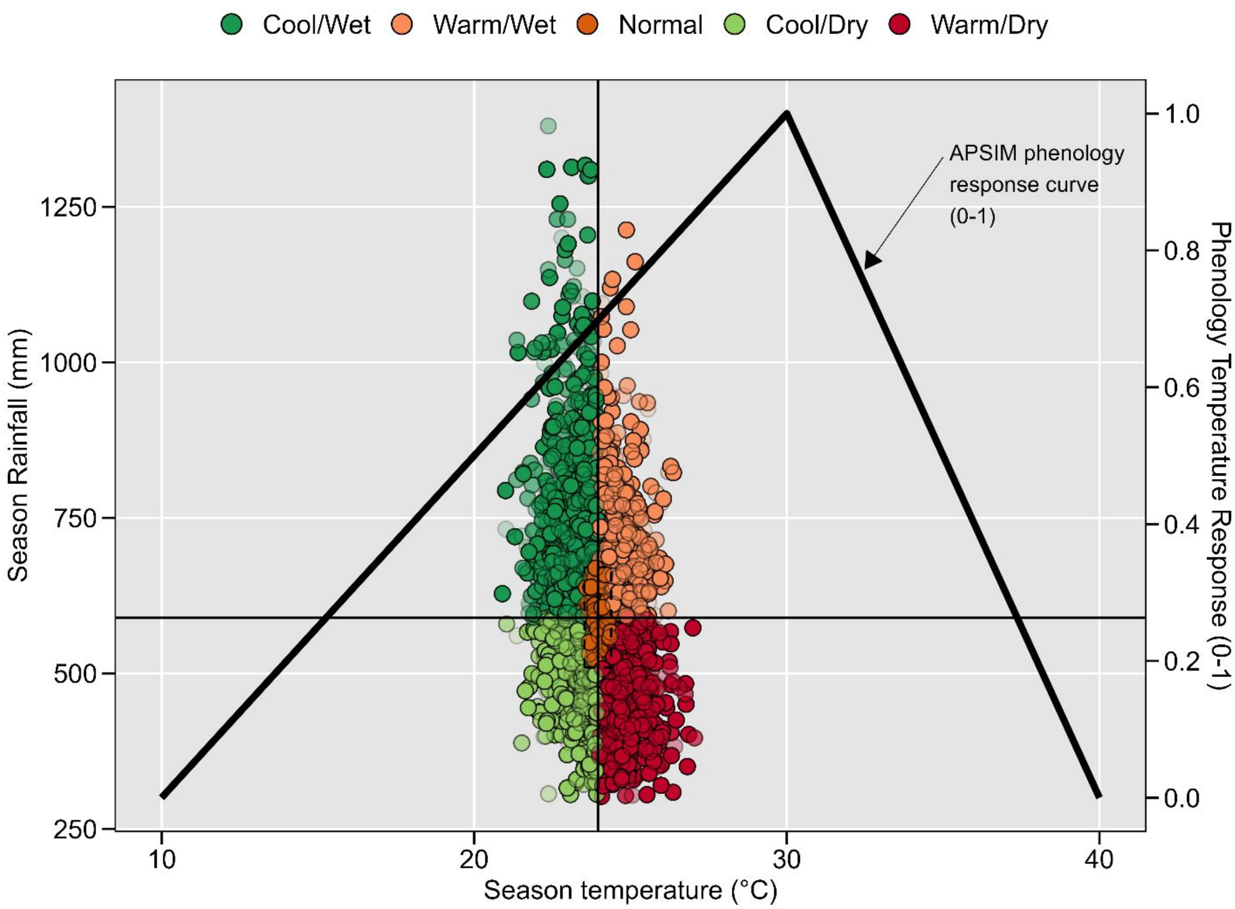
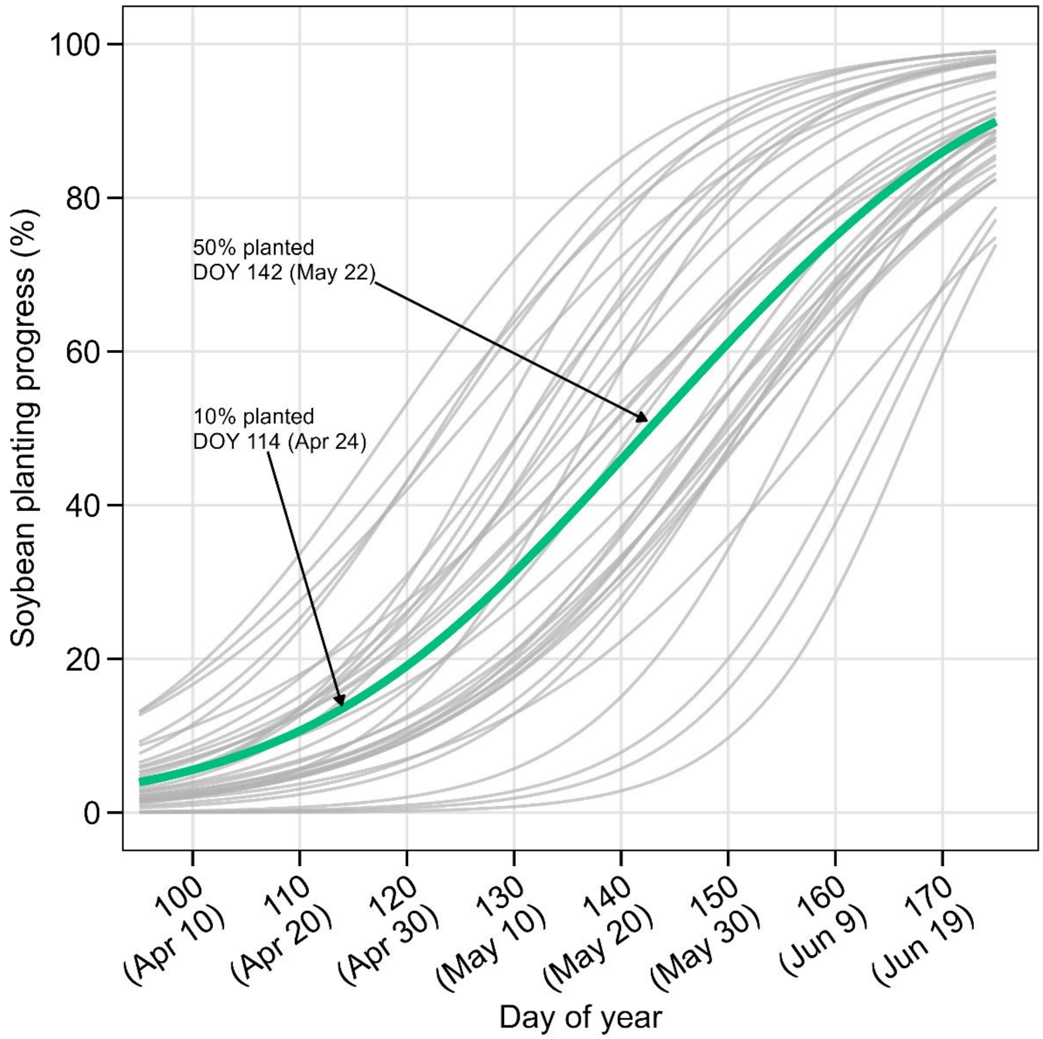


Figure S3. Environmental characterization of baseline conditions across all studied field-years. The APSIM phenology response curve (0-1) is presented, where 1 represents maximum development rate. The solid vertical and horizontal lines indicate the median growing-season average temperature and total growing-season precipitation, respectively.



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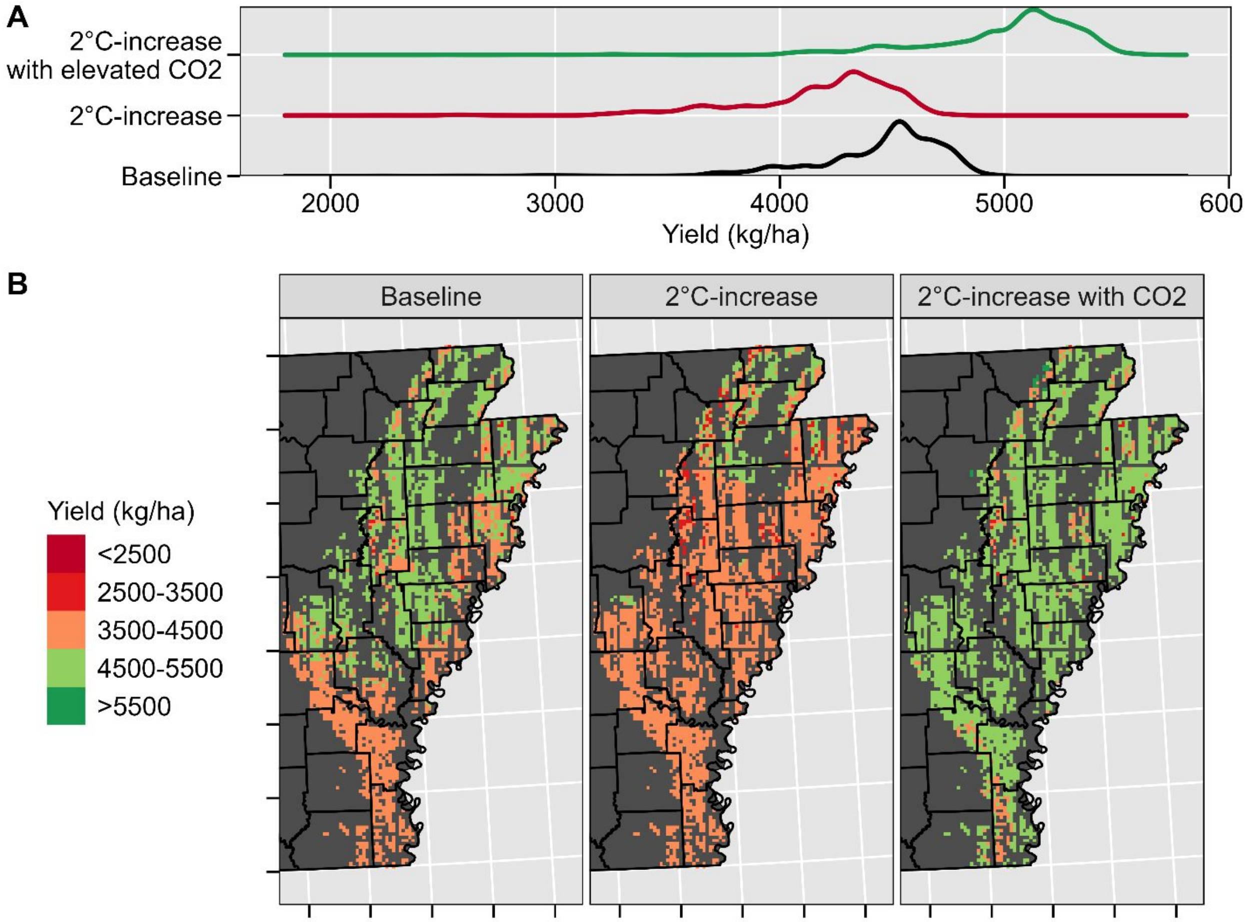
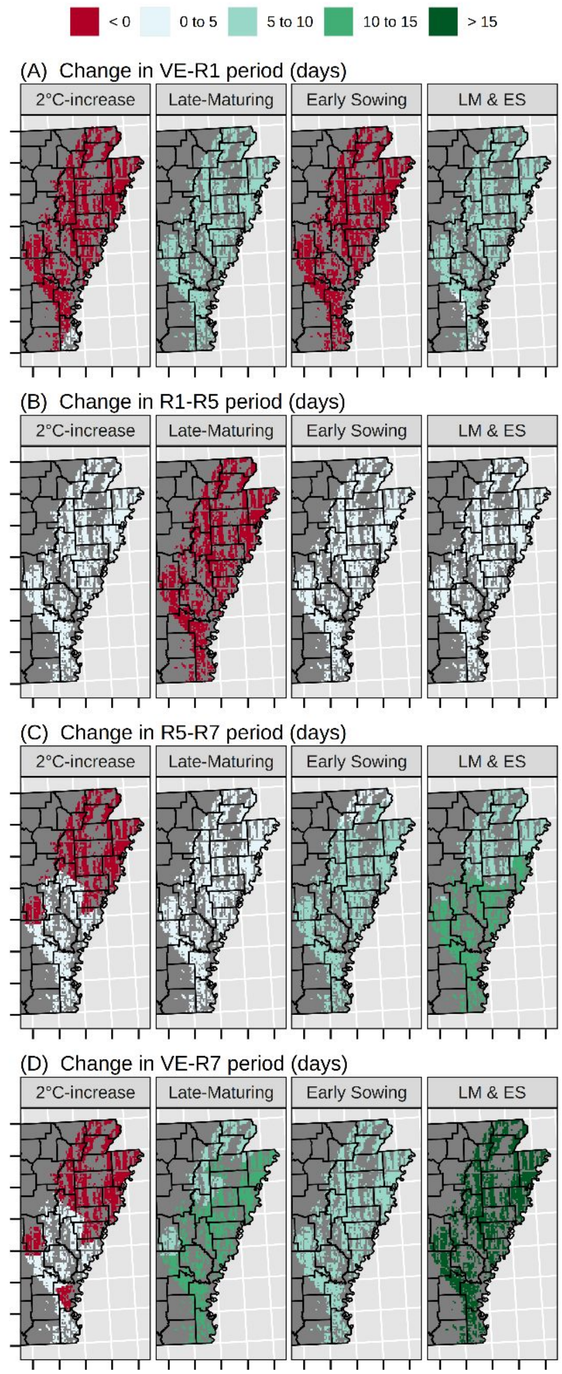


Figure S5. Density distribution of simulated soybean yields without adaptation under the baseline and +2°C scenarios with and without elevated [CO2] (A), and its geographical distribution (B).



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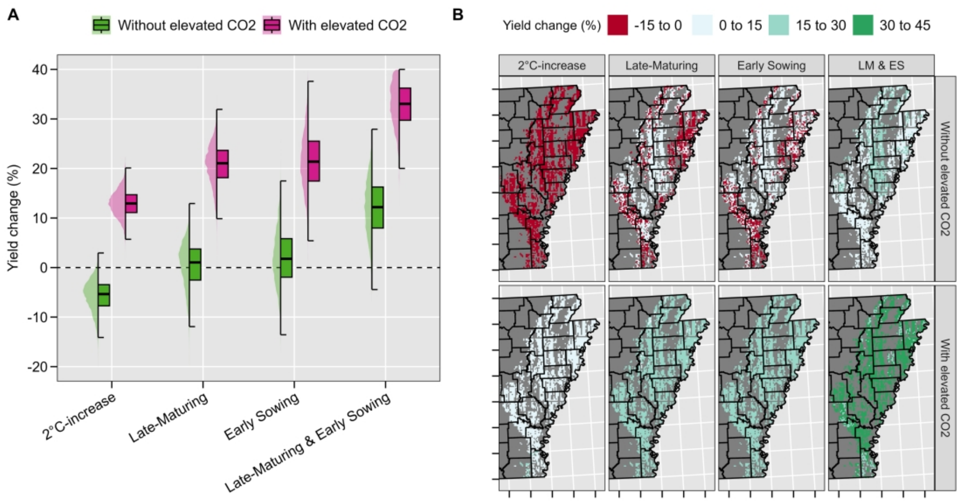

Figure 1. Overall relative effects of rising temperatures and adaptive genotypic and agronomic management strategies on soybean seed yield across 4,651 soybean fields and 40 weather years in the US Mid-South (A), and their geographical distribution (B). Values per pixel in panel B are averages across 40 weather years. LM & ES represent the combined effect of a late-maturing variety and early sowing. All adaptive scenarios were simulated under +2°C.

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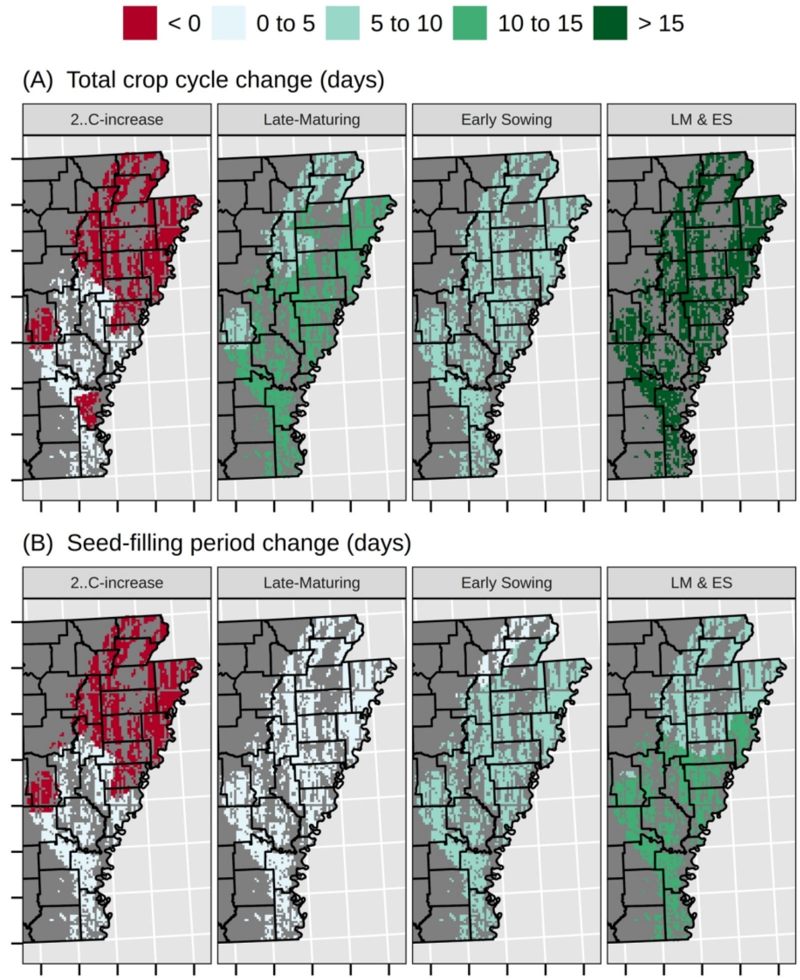

Figure 2. Geographical distribution of the relative effects of rising temperatures and adaptive genotypic and agronomic management strategies on changes in soybean total crop cycle (A) and seed-filling period (B) compared to baseline across 4,651 soybean fields in the US Mid-South. Values per pixel are averages across 40 weather years. LM & ES represent the combined effect of a late-maturing variety and early sowing. All adaptive scenarios were simulated under +2°C.

179x179mm (300 x 300 DPI)